# 1A, 200V - 1000V Fast Recovery Surface Mount Rectifier

## FEATURES

- Glass passivated chip junction
- Ideal for automated placement
- Low power loss, high efficiency
- Fast switching for high efficiency
- Low profile package
- Moisture sensitivity level: level 1, per J-STD-020
- RoHS Compliant
- Halogen-free

## APPLICATIONS

- DC to DC converter
- Switching mode converters and inverters
- General purpose

## MECHANICAL DATA

- Case: Thin SMA
- Molding compound meets UL 94V-0 flammability rating
- Terminal: Matte tin plated leads, solderable per J-STD-002
- Meet JESD 201 class 2 whisker test
- Polarity: Indicated by cathode band
- Weight: 0.029g (approximately)

| KEY PARAMETERS | | |
|---|---|---|
| **PARAMETER** | **VALUE** | **UNIT** |
| $I_F$ | 1 | A |
| $V_{RRM}$ | 200 - 1000 | V |
| $I_{FSM}$ | 30 | A |
| $T_{J\ MAX}$ | 175 | °C |
| Package | Thin SMA | |
| Configuration | Single die | |

Pb-Free RoHS COMPLIANT HALOGEN FREE

**Thin SMA**

| ABSOLUTE MAXIMUM RATINGS ($T_A$ = 25°C unless otherwise noted) | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **PARAMETER** | | **SYMBOL** | **RS1DAL** | **RS1GAL** | **RS1JAL** | **RS1KAL** | **RS1MAL** | **UNIT** |
| Marking code on the device | | | RS1DAL | RS1GAL | RS1JAL | RS1KAL | RS1MAL | |
| Repetitive peak reverse voltage | | $V_{RRM}$ | 200 | 400 | 600 | 800 | 1000 | V |
| Reverse voltage, total rms value | | $V_{R(RMS)}$ | 140 | 280 | 420 | 560 | 700 | V |
| Forward current | | $I_F$ | 1 | | | | | A |
| Surge peak forward current single half sine wave superimposed on rated load | t = 8.3ms | $I_{FSM}$ | 30 | | | | | A |
| | t = 1.0ms | | 100 | | | | | A |
| Junction temperature | | $T_J$ | -55 to +175 | | | | | °C |
| Storage temperature | | $T_{STG}$ | -55 to +175 | | | | | °C |

## THERMAL PERFORMANCE

| PARAMETER | SYMBOL | TYP | UNIT |
|---|---|---|---|
| Junction-to-lead thermal resistance | $R_{\Theta JL}$ | 19 | °C/W |
| Junction-to-ambient thermal resistance | $R_{\Theta JA}$ | 81 | °C/W |
| Junction-to-case thermal resistance | $R_{\Theta JC}$ | 19 | °C/W |

**Thermal Performance Note:** Units mounted on PCB (5mm x 5mm Cu pad test board)

## ELECTRICAL SPECIFICATIONS ($T_A$ = 25°C unless otherwise noted)

| PARAMETER | | CONDITIONS | SYMBOL | TYP | MAX | UNIT |
|---|---|---|---|---|---|---|
| Forward voltage[1] | RS1DAL RS1GAL RS1JAL | $I_F$ = 0.5A, $T_J$ = 25°C | $V_F$ | 0.90 | - | V |
| | | $I_F$ = 1.0A, $T_J$ = 25°C | | 0.97 | 1.30 | V |
| | | $I_F$ = 0.5A, $T_J$ = 125°C | | 0.75 | - | V |
| | | $I_F$ = 1.0A, $T_J$ = 125°C | | 0.83 | 0.94 | V |
| | RS1KAL RS1MAL | $I_F$ = 0.5A, $T_J$ = 25°C | | 0.96 | - | V |
| | | $I_F$ = 1.0A, $T_J$ = 25°C | | 1.04 | 1.30 | V |
| | | $I_F$ = 0.5A, $T_J$ = 125°C | | 0.80 | - | V |
| | | $I_F$ = 1.0A, $T_J$ = 125°C | | 0.90 | 1.11 | V |
| Reverse current @ rated $V_R$[2] | | $T_J$ = 25°C | $I_R$ | - | 1 | μA |
| | | $T_J$ = 125°C | | - | 33 | μA |
| Reverse recovery time | RS1DAL RS1GAL | $I_F$ = 0.5A, $I_R$ = 1.0A, $I_{rr}$ = 0.25A | $t_{rr}$ | - | 150 | ns |
| | RS1JAL | | | - | 250 | ns |
| | RS1KAL RS1MAL | | | - | 500 | ns |
| Junction capacitance | | 1MHz, $V_R$ = 4.0V | $C_J$ | 7 | - | pF |

**Notes:**

1. Pulse test with PW = 0.3ms
2. Pulse test with PW = 30ms

## ORDERING INFORMATION

| ORDERING CODE[1] | PACKAGE | PACKING |
|---|---|---|
| RS1xAL | Thin SMA | 14,000 / Tape & Reel |

**Notes:**

1. "x" defines voltage from 200V(RS1DAL) to 1000V(RS1MAL)

## CHARACTERISTICS CURVES

($T_A$ = 25°C unless otherwise noted)

**Fig.1 Forward Current Derating Curve**

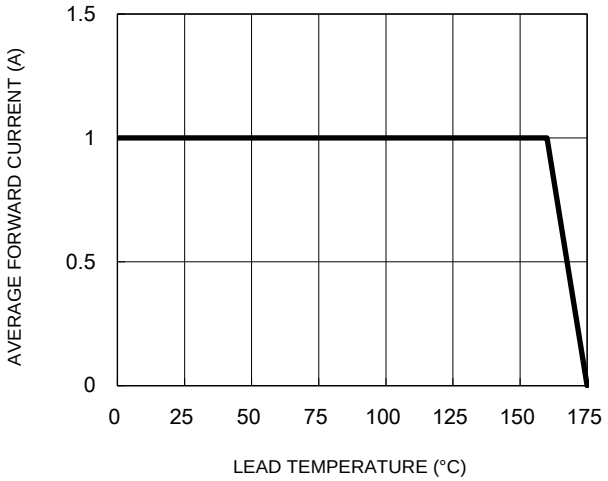



**Fig.2 Typical Junction Capacitance**

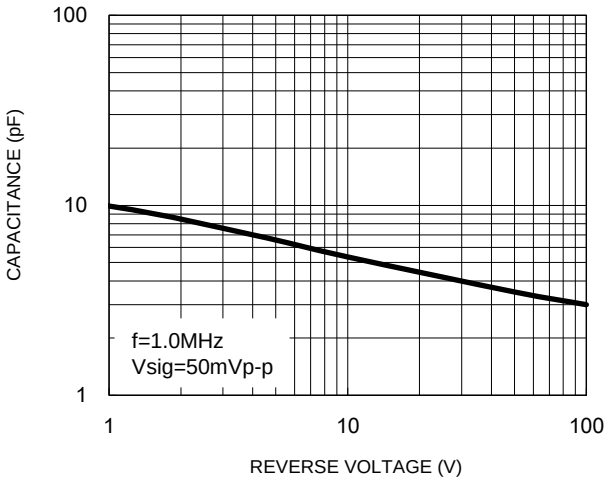



**Fig.3 Typical Reverse Characteristics**

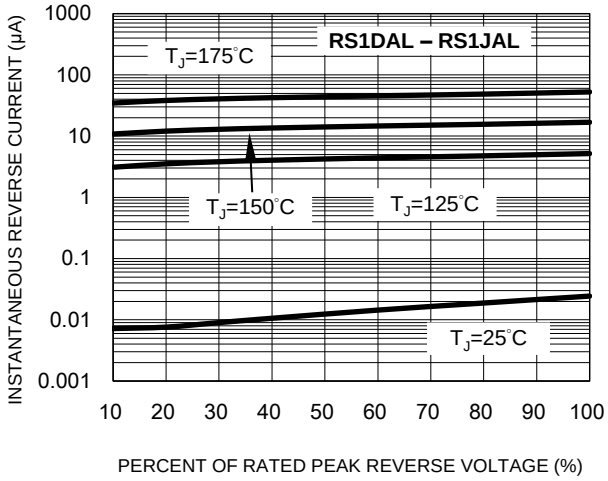



**Fig.4 Typical Forward Characteristics**

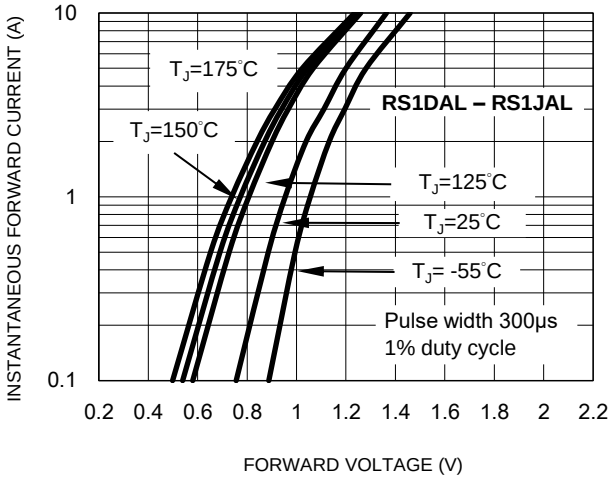



**Fig.5 Typical Reverse Characteristics**

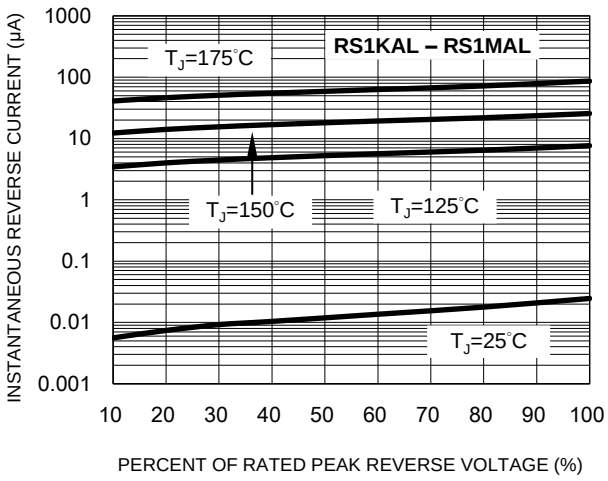
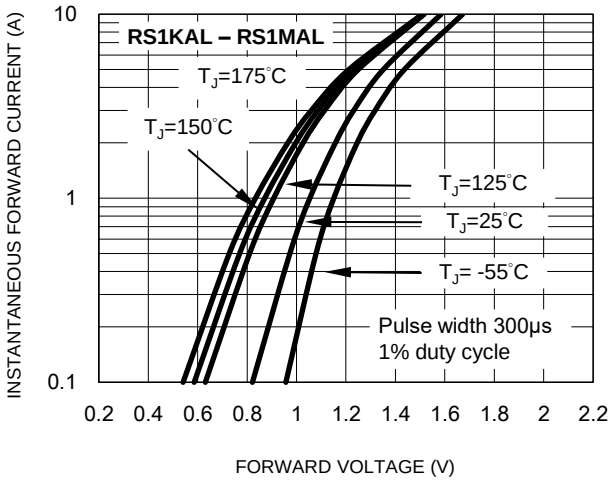


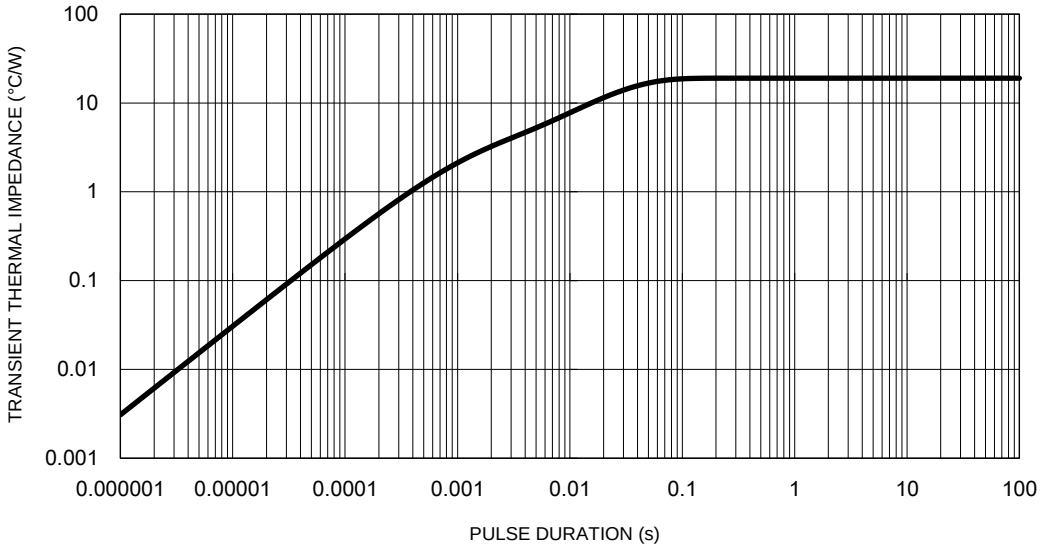

**Fig.6 Typical Forward Characteristics**



## CHARACTERISTICS CURVES

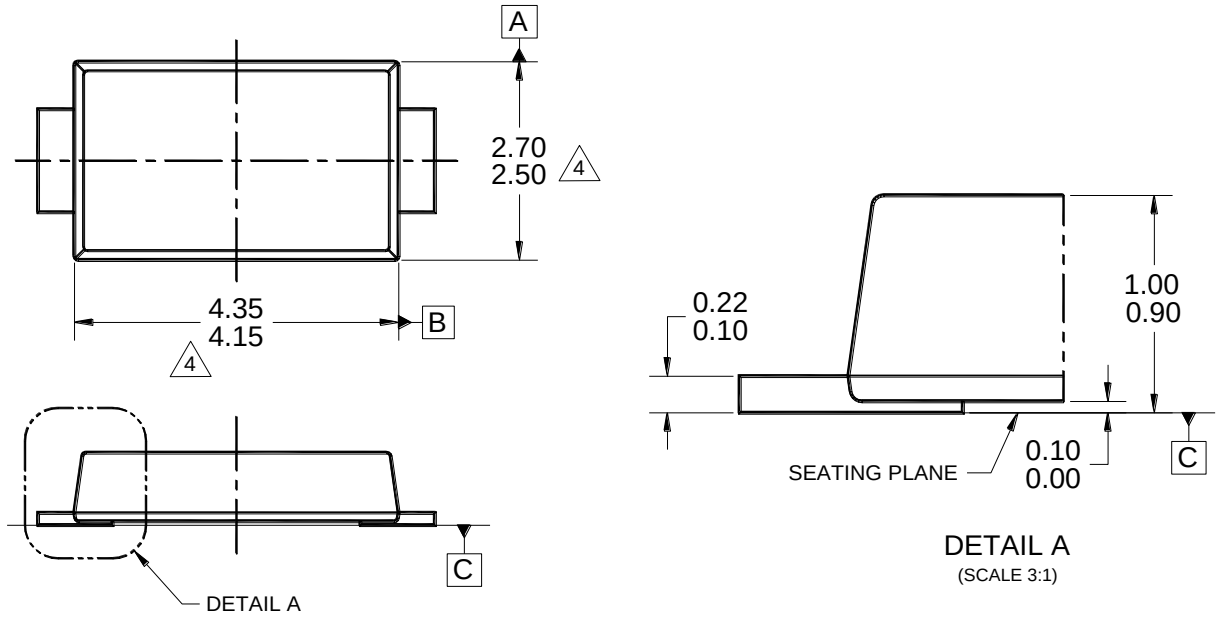
($T_A$ = 25°C unless otherwise noted)

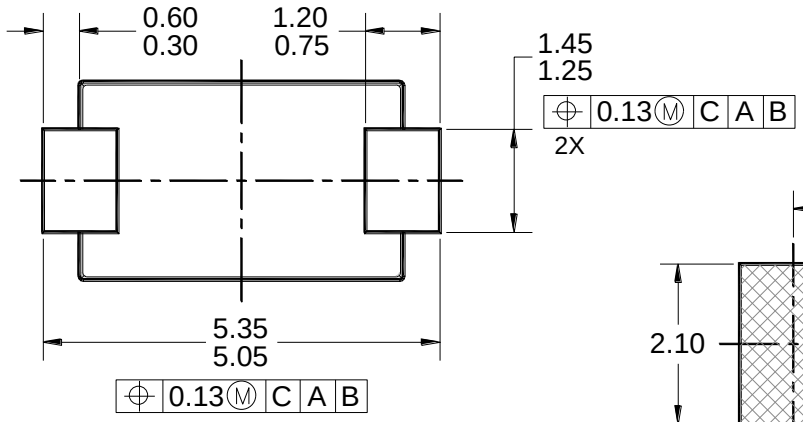
**Fig.7 Typical Transient Thermal Impedance**

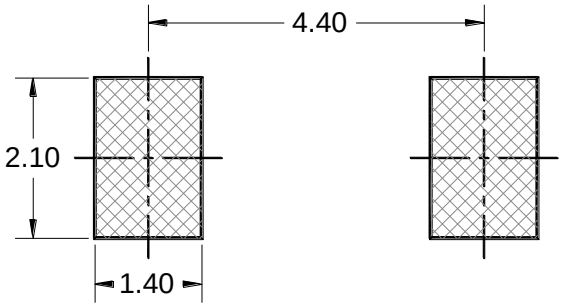


## PACKAGE OUTLINE DIMENSIONS

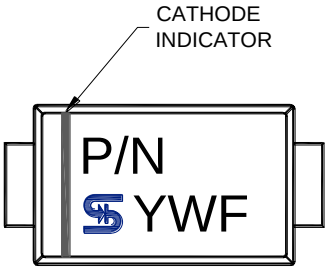
**Thin SMA**




SUGGESTED PAD LAYOUT




MARKING DIAGRAM

P/N = MARKING CODE
YW = DATE CODE
F = FACTORY CODE

NOTES: UNLESS OTHERWISE SPECIFIED

1. ALL DIMENSIONS ARE IN MILLIMETERS.
2. DIMENSIONING AND TOLERANCING PER ASME Y14.5M-2009.
3. PACKAGE OUTLINE REFERENCE: JEDEC DO-221, VARIATION AC, ISSUE B.
4. MODED PLASTIC BODY DIMENSIONS DO NOT INCLUDE MOLD FLASH.
5. SUGGESTED PAD LAYOUT IS FOR REFERENCE PURPOSE ONLY.
6. DWG NO. REF: HQ2SD07-TSMA-074 REV A.

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